

The Complexity of Clearing with Hard Budgets

Companion to *Prediction Markets Are Fisher Markets*

Draft — July 2026. One step of the hardness proof (Theorem★) awaits interval-arithmetic formalization and is marked as such throughout.

Abstract. The main paper shows that batch clearing of prediction-market limit orders under hard market-maker budget caps has a non-convex feasible set, and leaves its complexity as an open problem. This note settles most of it, in both directions. *Hidden convexity:* on books where each binary market carries at most one order, the substitution to expenditure coordinates makes the problem concave — polynomial-time solvable despite the non-convex feasible set; in particular the main paper’s non-convexity witness family is a coordinate artifact (Theorem 1). *The genuine obstruction* is resting retail liquidity: a sell wall makes the welfare–expenditure frontier non-concave, with the marginal welfare per dollar of MM capital jumping upward by the explicit factor $1 + (q^*/b)(1 - L_R)$ the moment MM demand reaches the wall — the inframarginal repricing tax vanishes exactly when the counterparty appears (Proposition 1). *Hardness:* an assembly of wall gadgets encodes SUBSET-SUM with exactly rational instance parameters; two welfare lemmas with explicit constants and a budget-window threshold scheme yield weak NP-hardness of the decision problem, modulo one interval-arithmetic formalization step (Theorem★). *Approximation stays easy:* the separation analysis forces the reduction’s yes/no gap to be exponentially small in value, so exact optimization is what is hard — the main paper’s convex relaxation, which lands inside the feasible set with a quantified additive gap, is the approximation algorithm one should run. The moral: budgets are benign until retail liquidity arrives in lumps — and even then, only exactness is hard.

1. Introduction

The main paper replaces the hard-budget clearing model by a convex one and argues the replacement is better economics. But the original question does not go away: how hard is the model everyone writes down first? Its feasible set is non-convex for every temperature $b > 0$ (main paper, §3), and no polynomial-time algorithm is known. Non-convexity, however, is not hardness — a warning made concrete by Devanur, Dudík, Huang, and Pennock (2015), who found hidden convexity in the sequential single-trader version of exactly this problem. Settling the question requires either a convexification or a reduction, and this note supplies most of both:

1. **Thin books are easy** (Theorem 1). If each binary market carries at most one order, the substitution from fills to capital *expenditures* turns the problem into a separable concave maximization over a simplex — polynomial time, for any number of market makers and any budgets. The main paper’s two-lobe witness family lives in this class: its celebrated non-convexity is a coordinate artifact, and building an NP-hardness gadget out of those lobes — the obvious plan — provably cannot work.
2. **Retail walls are the true obstruction** (Proposition 1). One resting sell order makes the per-market welfare–expenditure frontier genuinely non-concave, in every coordinate system: marginal welfare per dollar *jumps up* when MM demand reaches the wall, because the MM stops paying the inframarginal repricing tax exactly when

a counterparty appears. Non-convexity in this problem is not the market maker’s sigmoid; it is liquidity arriving in lumps.

3. **Hardness, one step from a theorem** (Theorem★, §5). Wall gadgets scaled by SUBSET-SUM weights, one shared budget, and a two-point Lagrangian argmax make “which walls to absorb” a subset-sum decision. The instance parameters are exactly rational; the welfare analysis is two short lemmas with explicit constants; a budget-window lemma handles the threshold. The single remaining step is interval-arithmetic bookkeeping for the derived constants, marked explicitly.
4. **Only exactness is hard** (§6). The same deviation analysis that proves separation also shows a no-instance recovers the welfare bound up to an exponentially small deficit. Any reduction of this type therefore has $2^{-\text{poly}}$ yes/no gaps: hardness lives entirely in the last bits of precision. Additive approximation remains easy — and the main paper’s risk-averse relaxation, which is feasible for the hard budgets with additive gap $\sum_k \Delta_k^2 / 2B_k$, is precisely the approximation algorithm the hardness result recommends.

Related work. The adjacent hardness results in market design are driven by indivisibilities: winner determination and core pricing in combinatorial exchanges with financially constrained buyers is Σ_2^P -hard (Bichler and Waldherr 2018), and assignment markets with budgets are NP-hard even under complete information (Batziou, Bichler, and Fichtl 2022). Here everything is divisible; the hardness mechanism is the bilinear price-times-fill budget coupling, which is disjoint from packing. The #P-hardness of combinatorial LMSR pricing (Chen et al. 2008) is orthogonal in the same sense: state-space explosion rather than budget coupling. In expenditure coordinates our problem is the maximization of a sum of non-concave scalar functions over a knapsack — the genre of sigmoidal programming, known NP-hard in general (Udell and Boyd 2014); the contribution here is that clearing *reaches* that genre only through retail walls (Theorem 1 says the MM-only fragment does not), and that within it, an exact reduction with rational instances can be assembled.

2. The Decision Problem

We use the main paper’s setting, specialized to n independent binary market groups. Orders have limit prices, sides, caps; fills are $\mathbf{q} \in [0, \overline{\mathbf{Q}}]$; group net demands $\mathbf{D}(\mathbf{q})$ are linear in fills; the smoothed minting cost is $C_b(\mathbf{D}) = \sum_g b \ln(e^{D_{vg}/b} + e^{D_{ng}/b})$ and clearing prices are its gradient. Risk-neutral welfare is $F(\mathbf{q}) = \sum_i w_i q_i - C_b(\mathbf{D}(\mathbf{q}))$. Market maker k ’s capital consumption at the clearing prices is $h_k(\mathbf{q}) = \sum_{i \in \text{MM}_k} p_{\omega(i)}(\mathbf{q}) q_i$.

Problem (D-HBC). *Given rational order data, temperatures, budgets B_k , and a rational threshold W : does there exist a UCP-valid $\mathbf{q} \in [0, \overline{\mathbf{Q}}]$ with $h_k(\mathbf{q}) \leq B_k$ for all k and $F(\mathbf{q}) \geq W$?*

Two modeling commitments. *UCP validity* requires fills to respect limit prices at the clearing prices (a sell fills only at $p \geq L$); without it, the optimizer may overfill a resting sell below its limit to depress prices and relax the MM’s budget — a welfare-losing but budget-saving move the honest model forbids. The wall mechanism of §4 survives either choice (the kink sharpens without UCP), but the reduction commits to the UCP-valid variant. *Weak hardness* is the natural target: the non-convexity is metric rather than combinatorial, the adjacent strong-hardness results all lean on indivisibilities, and §6 shows the yes/no gaps of our reduction are necessarily tiny in value — all three point the same way.

3. Hidden Convexity on Thin Books

Call a book *thin* if each binary market carries at most one order — necessarily an MM buy order (retail orders with locked capital are unaffected by budgets; relabel sells as buys of the complement). The main paper’s non-convexity witness is a thin book: one MM, one order per market, capital consumption $h(\mathbf{q}) = \sum_j f_b(q_j)$ with $f_b(q) = \sigma(q/b)q$, whose sublevel sets have two-lobe geometry.

Theorem 1 (Thin Books Convexify). *On a thin book, substitute per-market expenditure $e_j = f_{b_j}(q_j)$ (monotone, hence bijective). The clearing problem becomes*

$$\max \sum_j \gamma_j(e_j) \quad \text{s.t.} \quad \sum_{j \in \text{MM}_k} e_j \leq B_k \quad \forall k, \quad e_j \in [0, f_{b_j}(\bar{Q}_j)]$$

where $\gamma_j = \varphi_j \circ f_{b_j}^{-1}$ and $\varphi_j(q) = L_j q - C_{b_j}(q) + C_{b_j}(0)$ is market j ’s welfare ($L_j \in (1/2, 1]$; below $1/2$ the order never fills). Each γ_j is **strictly concave**, so the substituted problem is a separable concave maximization over box-and-simplex constraints: D-HBC on thin books is polynomial-time solvable, for any number of MMs.

Proof. Markets partition across MMs and each budget constraint is separable, so it suffices to prove concavity of one γ . Take $b = 1$ by the scaling identity $f_b(bt) = b f_1(t)$ and parametrize by q on the active domain $\{\sigma(q) < L\}$: the frontier slope is $\gamma' = (L - \sigma)/(\sigma(1 + q(1 - \sigma)))$. Using $\sigma' = \sigma(1 - \sigma)$, a direct computation (verified symbolically; `scripts/hardness_debts.py`) gives

$$\frac{d}{dq} \ln \gamma' = -(1 - \sigma) \left[\underbrace{\frac{\sigma}{L - \sigma} + \frac{2 + q(1 - 2\sigma)}{1 + q(1 - \sigma)}}_{=: \mathcal{B}(q, L)} \right]$$

For $L = 1$, multiplying $\mathcal{B}$ by the positive quantity $(1 - \sigma)(1 + q(1 - \sigma))$ yields exactly $(2 - \sigma) + q(1 - \sigma)^2 > 0$. For $L < 1$, $\mathcal{B}$ only increases, since $\sigma/(L - \sigma) > \sigma/(1 - \sigma)$ on the active domain. So γ' is strictly decreasing in q ; since e is strictly increasing in q , γ is strictly concave in e . $\square$

Note what the proof is *not*: an inheritance. The expenditure derivative f' is non-monotone (it rises to ≈ 1.1 at the inflection $q \approx 2.4b$ before falling back to 1); the concavity of γ is a cancellation between the falling marginal welfare and the non-monotone marginal cost — which is presumably why the witness family was mistaken for a source of hardness. Two consequences. First, the main paper’s §3 witness shows the feasible set is non-convex *as written*, and nothing more: the associated optimization problem is easy. Second, any hardness construction must leave the thin class — it needs either retail counterparties or several MM orders per market. The next section shows the first option suffices.

4. The Wall Mechanism

Add one resting retail sell-YES order (limit L_R , size Q_R) to a binary market where the MM buys YES (limit 1, cap on the wall as in §5). Under UCP-valid clearing, as the MM’s fill q grows with $q^* = b \ln(L_R/(1 - L_R))$: *pre-wall* ($\sigma(q/b) < L_R$), retail cannot fill and $e(q) = \sigma(q/b)q$; *on the wall*, the price pins at L_R , retail absorbs $r = q - q^*$, and expenditure is linear, $e = L_R q$, while welfare gains $1 - L_R$ per unit.

Proposition 1 (The Frontier Kinks Upward at the Wall). *The welfare-expenditure frontier $\gamma(e)$ of this market is non-concave: its slope jumps up at the wall,*

$$\gamma'(e^* -) = \frac{1 - L_R}{L_R [1 + (q^*/b)(1 - L_R)]} \rightarrow \gamma'(e^* +) = \frac{1 - L_R}{L_R}$$

a jump by the factor $1 + (q^/b)(1 - L_R) > 1$. (At $b = 1$, $L_R = 0.7$: slopes $0.342 \rightarrow 0.429$.)*

Proof. Pre-wall, $de/dq = \sigma + (q/b)\sigma(1 - \sigma)$ and $d\varphi/dq = 1 - \sigma$; evaluate at $\sigma = L_R$. On the wall, $de/dq = L_R$ and $d\varphi/dq = 1 - L_R$ exactly. Expenditure is strictly increasing in q throughout, so the slope discontinuity in q is a slope discontinuity in e . $\square$

The economics deserves the emphasis. Pre-wall, a marginal share costs $p + q dp$: the MM re-prices its entire inframarginal position against itself. On the wall, retail supply pins the price, $dp = 0$, and the repricing tax vanishes — *exactly when the counterparty appears*, marginal welfare per dollar improves discontinuously. The non-convexity of hard-budget clearing is not the market maker's own price impact; it is the lumpiness of resting liquidity. (Without the UCP constraint the kink survives and sharpens: overfilling the wall buys expenditure reductions at second-order welfare cost, flattening the frontier to the left of the wall, so the slope jump is from ≈ 0 to $(1 - L_R)/L_R$.)

5. The Reduction

Construction. Given SUBSET-SUM data — positive integers $a_1, \dots, a_n$, target T , $A := \sum_j a_j$ — build n independent binary markets sharing one MM. Market j is the wall gadget *dilated by a_j* : temperature $b_j = a_j$, retail wall at $L_R = 7/10$ of size $6a_j$, MM buy order at limit 1 capped at $\bar{q}_j = 6a_j$. By the scaling identity $f_b(bt) = bf_1(t)$, market j 's frontier is an exact dilation, $\gamma_j(e) = a_j \gamma(e/a_j)$, of one reference frontier γ . Two design points matter. The cap sits at a rational point *on* the wall ($\hat{q} = 6 < q^* + Q_R$, so the wall never exhausts): the frontier ends at $\hat{x}^+ = L_R \hat{q} = 21/5$ — rational — and the shape is a strictly concave arc (Theorem 1's proof applies verbatim pre-wall) followed by a linear segment of strictly higher slope. Consequently *every instance parameter is rational*. The transcendental quantities live only in derived constants, which enter through the budget and threshold — our free choices.

The supporting line. Let x^- be the tangency point where the chord to the endpoint $\hat{x}^+$ touches the arc, μ^* the chord slope, $s = \hat{x}^+ - x^-$, and

$$\bar{G}(x) = \gamma(x^-) + \mu^*(x - x^-) - \gamma(x)$$

the gap to the supporting line. Strict concavity off the wall and linearity (with slope $\neq \mu^*$) on it give $\bar{G} \geq 0$ with zeros *exactly* $\{x^-, \hat{x}^+\}$. Write lobe coordinates $y_j = (e_j/a_j - x^-)/s$; the two lobes are $y \in [0, 1]$. Reference constants (rational gadget): $x^- = 0.42766$, $\mu^* = 0.42651$, $s = 3.77234$; $\kappa_2 := (\text{wall slope} - \mu^*)s = 7.783 \times 10^{-3}$; and $\kappa_1 := \inf_{y \neq 0} \bar{G}(x^- + ys)/y^2 \geq (s^2/2) \min_{\text{arc}}(-\gamma'') \geq 3.04$ by second-order Taylor, since $\bar{G}(x^-) = \bar{G}'(x^-) = 0$ and $\bar{G}'' = -\gamma''$ has the explicit form $R(1 - \sigma)\mathcal{B}/f'$ from Theorem 1's proof, with arc minimum 0.4271 at the wall start.

Lemma 1 (Welfare Decomposition). *For any feasible UCP-valid clearing with per-market MM expenditures $e_j \in [0, a_j \hat{x}^+]$ and $\sum_j e_j \leq B$,*

$$F \leq \text{Bound}(B) - \mu^* \left(B - \sum_j e_j \right) - \sum_j a_j \bar{G}(e_j/a_j), \quad \text{Bound}(B) := A\gamma(x^-) + \mu^*(B - Ax^-)$$

with equality on frontier configurations; in particular $F = \text{Bound}$ exactly on full-budget lobe configurations.

Proof. Market j 's welfare is at most $a_j \gamma(e_j/a_j)$ (frontier definition plus dilation), and $\gamma(x) = [\gamma(x^-) + \mu^*(x - x^-)] - \bar{G}(x)$ pointwise. Sum over j ; bound $\sum e_j$ by B . $\square$

Lemma 2 (Separation). *Set $c_0 = \frac{1}{8} \min(\kappa_2, \kappa_1/A)$ and the ideal budget $B_0 = Ax^- + Ts$. For any budget $B \in [B_0, B_0 + c_0/(2\mu^*)]$:*

- (YES) *if some subset sums to T , its lobe configuration is feasible and achieves $F = X := A\gamma(x^-) + \mu^*Ts$;*
- (NO) *if no subset sums to T , every feasible point has $F \leq X + \mu^*(B - B_0) - c_0$.*

Proof. YES: the configuration spends $B_0 \leq B$ and every $\bar{G}$ term vanishes. NO: let $u = B - \sum e_j \geq 0$. If $u \geq s/2$, Lemma 1 already yields a deficit $\mu^*s/2 \gg c_0$. Otherwise round each y_j to the nearest of $\{0, 1\}$: the rounded configuration is a subset sum $s' \neq T$, so $\sum_j a_j d_j \geq |T - s'| - (u + B - B_0)/s \geq 1/2$, where d_j is the rounding distance. Each market's penalty obeys $\bar{G} \geq \kappa_2 d_j$ for wall-side deviations and $\bar{G} \geq \kappa_1 d_j^2$ off-wall. Splitting the mass $\sum a_j d_j \geq 1/2$ between regimes: the linear regime pays at least $\kappa_2/4$, or the quadratic regime carries mass $\geq 1/4$ and — optimally spread across markets — pays at least $\kappa_1/(16A)$. Either way the penalty exceeds c_0 , with slack absorbing the budget enlargement. $\square$

Lemma 3 (Budget Window). *Pick any rational $B \in [B_0, B_0 + c_0/(2\mu^*)]$ and rational $W^* \in (X - c_0/4, X)$. Then YES instances of SUBSET-SUM map to YES instances of D-HBC and NO to NO. Both B and W^* require $O(\log(A/c_0))$ bits, i.e. polynomially many.*

Proof. YES: $F \geq X > W^*$ by Lemma 2. NO: $F \leq X + \mu^*(B - B_0) - c_0 \leq X - c_0/2 < W^*$. $\square$

Theorem★. D-HBC — UCP-valid batch clearing with hard budgets and binary-encoded rational inputs — is (weakly) NP-hard.

What the star covers. The reduction is the construction above with Lemma 3's budget and threshold. The instance parameters are exactly rational; the constants $x^-, \mu^*, \gamma(x^-), \kappa_1, \kappa_2$ entering B and W^* are roots and values of explicit logistic-log equations with concrete transversality margins (μ^* clears the wall slope by 2.1×10^{-3} ; the tangency is interior by 0.42 in q), so interval Newton computes them to any polynomial number of bits in polynomial time; and the Taylor certificate for κ_1 needs a grid-plus-Lipschitz (or interval) evaluation of the explicit $-\gamma''$. Writing these three routine verifications down carefully is all that separates the star from a theorem. Nothing conceptual is open.

Numerical validation. The pipeline was validated end-to-end on live instances ($a = (3, 5, 7)$): the YES instance $T = 8$ accepts with margin $+4.9 \times 10^{-4}$, landing exactly on the predicted lobes; the NO instance $T = 9$ rejects with margin -7.3×10^{-3} , and a 400-start global search escapes through precisely the integer-1 wall-back route that Lemma 2 prices, with measured deficit matching κ_2 to 0.15% (`scripts/hardness_rational.py`, `scripts/hardness_e2e.py`).

6. Discussion

Only exactness is hard. The separation analysis cuts both ways. A NO instance recovers the welfare bound up to $\Theta(\kappa_1/A)$ — exponentially small in the input length whenever the subset-sum weights are — by spreading tiny off-lobe deviations across markets. So the yes/no gap of this reduction (and, we expect, of any reduction from a number problem) is necessarily $2^{-\text{poly}}$: the hardness of hard-budget clearing lives entirely in the last bits of precision. Additive approximation is unobstructed, and the main paper already supplies the algorithm: its risk-averse convex relaxation is feasible for the hard budgets with additive welfare gap at most $\sum_k \Delta_k^2/2B_k$. We conjecture an additive

FPTAS exists for D-HBC; the moral stands regardless — *the convex relaxation is the right thing to run*, now with a hardness result explaining why nothing exact should be attempted.

Open questions. (1) *Strong hardness.* Our reduction inherits weakness from SUBSET-SUM; whether D-HBC is strongly NP-hard (or pseudo-polynomial) is open, and the tiny-gap phenomenon suggests genuine obstacles to strong hardness for the welfare-threshold version. (2) *The approximation frontier.* Additive FPTAS, or even a PTAS in relative terms near the LP scale? (3) *Ladders.* Between thin books (Theorem 1, easy) and retail walls (Proposition 1, hard) sits the retail-free regime with several MM orders per market. The expenditure substitution no longer separates; whether ladders alone can simulate a wall — an MM’s own resting orders creating the lumpy-liquidity kink for *another* MM — is the sharpest open boundary question, and would determine whether hardness requires retail at all. (4) *The UCP-free variant.* The kink survives without UCP validity (§4), but the frontier geometry differs; the reduction should transfer with modified constants.

Relation to the main paper. Its §3 exhibits the two-lobe witness; this note shows the witness’s non-convexity is a coordinate artifact (Theorem 1) while retail walls provide the load-bearing obstruction (Proposition 1) — a strictly sharper story that the main paper’s Open Problem 4 now summarizes in one line. Nothing in the main paper changes: its response to the obstacle was never “solve the non-convex problem” but “replace the model and approximate the original with a guarantee,” and the results here say that response was the right one — exactly, not just heuristically.

*Companion to Prediction Markets Are Fisher Markets (2026); notation and the clearing model are defined there. Numerics and symbolic verifications: `scripts/hardness_gamma.py`, `hardness_debts.py`, `hardness_e2e.py`, `hardness_rational.py`. References: Batziou, E., Bichler, M., and Fichtl, M. (2022), Assignment markets with budget constraints, *arXiv:2205.06132*; Bichler, M. and Waldherr, S. (2018), Competitive equilibria in combinatorial exchanges with financially constrained buyers: computational hardness and algorithmic solutions, *arXiv:1807.08253*; Chen, Y., Fortnow, L., Lambert, N., Pennock, D. M., and Wortman, J. (2008), Complexity of combinatorial market makers, *EC*; Devanur, N. R., Dudík, M., Huang, Z., and Pennock, D. M. (2015), Budget constraints in prediction markets, *UAI*; Udell, M. and Boyd, S. (2014), Maximizing a sum of sigmoids, manuscript, Stanford University.*